SIMILARITY OF TRIANGLES AND POLYGONS

How to read the figures. —— given - - - auxiliary (added in the proof)
—— what is being proved or asked

9–1 INTRODUCTION

In this chapter we study relations between two polygons that have "the same shape." (Later on we explain precisely what we mean by "the same shape," and say that such figures are *similar to each other*.) Then we prove some theorems about similar triangles, and using these theorems we prove the Pythagorean theorem, which states that if a and b are lengths of the sides of a right triangle, and c is the length of the hypotenuse, then $c^2 = a^2 + b^2$. We see as we study this chapter, that the theorems on similar triangles depend very much on the properties of parallel lines. For this reason, we first continue our study of parallel lines.

9–2 TRANSVERSALS

The results of this chapter are based on the important theorem about parallel lines given below. In the theorem, one of the lines is *between* the other two. You should be able to tell what this means in terms of the ideas of betweenness that you have studied.

Exercise 9–1

If l, l', and l'' are three parallel lines, what should it mean to say that l' is between l and l''?

Theorem 9–1. *If l, l', and l'' are parallel lines, and m is a transversal intersecting the lines in points A, B, and C, respectively, such that $\overline{AB} \cong \overline{BC}$, and if m' is any other transversal intersecting l, l', and l'' in A', B', and C', respectively, then $\overline{A'B'} \cong \overline{B'C'}$.*

Remark. A brief way of stating the theorem is to say that if parallel lines cut off congruent segments on one transversal, then they cut off congruent segments on every transversal.

Proof.

STATEMENTS	**REASONS**
1. Consider lines m'' and m''' through A and B, respectively, and parallel to m' in Fig. 9–1.	Which postulate?
2. m'' intersects l' in D, and m''' intersects l'' in E.	Why?
3. A, A', B', and D are vertices of a parallelogram, as also are B, B', C', and E.	Since l, l', and l'' are parallel, as also are m', m'', and m'''.
4. $\overline{AD} \cong \overline{A'B'}$, $\overline{BE} \cong \overline{B'C'}$.	Why?
5. $\angle ABD \cong \angle BCE$, $\angle BAD \cong \angle CBE$.	Why?
6. $\overline{AB} \cong \overline{BC}$.	Given.
7. $\triangle ABD \cong \triangle BCE$.	A.S.A.
8. $\overline{AD} \cong \overline{BE}$.	Statement 7.
9. $\overline{A'B'} \cong \overline{B'C'}$.	Statements 4 and 8 and properties of congruence.

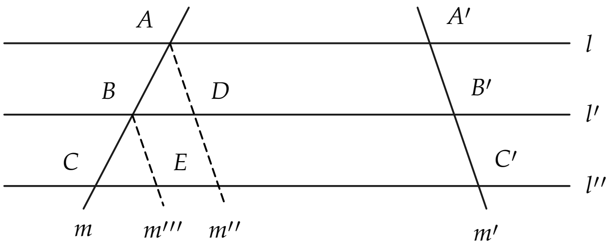

FIGURE 9–1

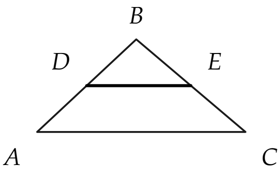

FIGURE 9–2

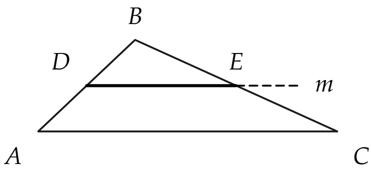

FIGURE 9–3

Corollary 9–1–1. *If a line is parallel to one side of a triangle and bisects another side, then it bisects the third side.*

Corollary 9–1–2. *If a line bisects two sides of a triangle, it is parallel to the third side, and the segment formed is half as long as the third side. (In Fig. 9–2,* $DE \parallel AC$*, and* $\overline{DE} = \frac{1}{2}\overline{AC}$*.)*

Proof. It is given that $\overline{AD} = \overline{DB}$, and $\overline{BE} = \overline{EC}$. Through D, let m be a line parallel to AC, as in Fig. 9–3. By Corollary 9–1–1, m meets BC in a point F, which bisects BC. Since a segment has only one bisector, $F = E$. To see that $\overline{DE} = \frac{1}{2}\overline{AC}$, consider a line m' through E, parallel to AB, as in Fig. 9–4. Then m' meets AC in G, which bisects AC. It is left for the student to show that $\overline{AC} = 2\overline{DE}$.

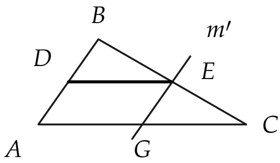

FIGURE 9–4

Exercise Group 9–2

1. The three sides of a triangle have lengths 5, 12, and 13. What are the lengths of the segments joining the midpoints of the sides?
2. In Fig. 9–5, $FC \cong CG$, $CD \parallel EG$, and $DG \parallel EF$. Prove that $\overline{EG} = 2\overline{CD}$.
3. Prove Corollary 9–1–1.
4. In Fig. 9–6, if $\overline{AB} = 2\overline{BC}$, how are $\overline{A'B'}$ and $\overline{B'C'}$ related? Suppose that $\overline{AB} = 3\overline{BC}$; $3\overline{AB} = 5\overline{BC}$. Prove your statements.
5. In Fig. 9–6, suppose that $\overline{AC} = 10$ in., $\overline{BC} = 2$ in., and $\overline{B'C'} = 3$ in. How long is $A'B'$?
6. In Fig. 9–6, if $\overline{AB} = 6$ in., $\overline{A'B'} = 8$ in., and $\overline{AC} = 12$ in., what is $\overline{B'C'}$?
7. In Fig. 9–7, if $\overline{EF} = 2$, $\overline{FG} = 3$, and $\overline{EP} = 3$, what is $\overline{EQ}$?
8. In the proof of Theorem 9–1, we assumed that m was not parallel to m'. Prove Theorem 9–1 if $m \parallel m'$.

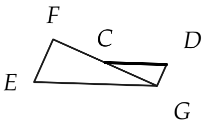

FIGURE 9–5

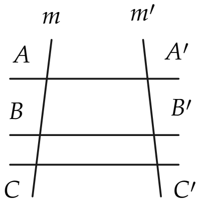

FIGURE 9–6

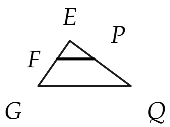

FIGURE 9–7

9–3 A CONSTRUCTION

In this section we show how to divide a segment into any number of congruent segments. The construction can be made with ruler and compass.

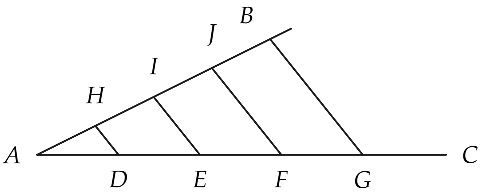

FIGURE 9–8

Construction 9–1. Given the segment AB, the problem is to divide it into a number of congruent segments. (In Fig. 9–8, the number is four.)

STATEMENTS	REASONS
1. Draw a ray AC, forming an angle with AB that is not a straight angle.	How do you know that such a ray exists?
2. With a compass, mark off points D, E, F, and G, in order $ADEFG$, such that $\overline{AD} \cong \overline{DE} \cong \overline{EF} \cong \overline{FG}$.	This can be done because of the congruence postulate and betweenness postulates.
3. Draw BG.	Postulate I.
4. Draw lines through D, E, and F, $\parallel BG$.	These lines exist by Theorem 8–1. The construction is RC. Why?
5. The parallel lines through D, E, and F intersect AB in points H, I, and J, respectively.	Why?
6. $\overline{AH} \cong \overline{HI} \cong \overline{IJ} \cong \overline{JB}$.	Theorem 9–1 on transversals.

Remark. In discussing measurement in Chapter 5, we had to divide the unit segment into tenths, and then hundredths, etc. The above construction shows that, with the parallel postulate, such a division of the unit segment is possible. At no time between Chapter 5 and now have we needed measurement for the development of our geometry.

Exercise Group 9–3

1. Draw a segment, and divide it into three segments of equal length. This is called *trisecting* a segment.

2. Draw a segment, and divide it into five congruent segments.

3. In Fig. 9–8, suppose that points I and J have been found. State two different ways to find H.

4. Can you divide a segment into four equal parts without using the above construction? eight equal parts? Is this an RC construction?

5. If, in Fig. 9–9, $\overline{AB} = 10$, $\overline{AD} = 2$, and $\overline{EC} = 3$, what is $\overline{BC}$?

6. If, in Fig. 9–9, $\overline{AD} = 6$, $\overline{DB} = 8$, and $\overline{EC} = 2$, what is $\overline{BC}$?

7. In Fig. 9–10, $\overline{AB} = 18$, $\overline{BC} = 3$, and $\overline{AD} = 24$. What is $\overline{EF}$?

8. Using Fig. 9–10, find $\overline{EF}$ if $\overline{BC} = 1$, $\overline{AB} = 2$, and $\overline{AD} = 3$.

9. In Fig. 9–11, if $BE \parallel CF \parallel DG$, and the lengths are as shown, how long is AG?

10. Solve Exercise 9 if everything is the same except that $\overline{AE} = 6$.

11. In Fig. 9–12, how long is AB?

*12. In Fig. 9–13, $ABCD$ is a parallelogram, and E and F are midpoints of AB and CD, respectively. Prove that AC intersects ED and BF in G and H, and that G and H trisect AC. Prove also that $\overline{EG} = \frac{1}{3}\overline{ED}$.

*13. In Fig. 9–14, x, y, z, and 1 refer to the lengths of the segments. If x is a whole number, show that $z = xy$.

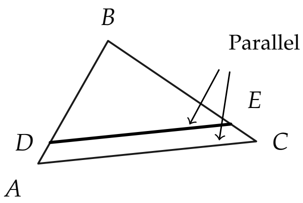

FIGURE 9–9

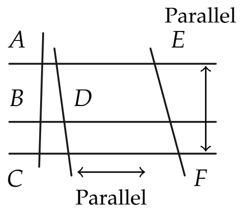

FIGURE 9–10

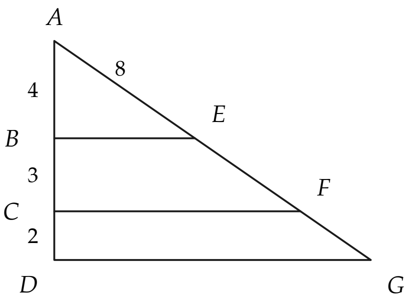

FIGURE 9–11

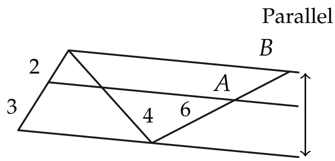

FIGURE 9–12

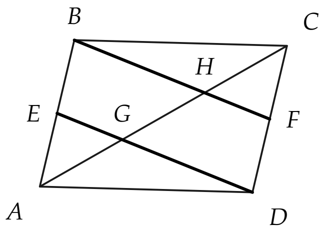

FIGURE 9–13

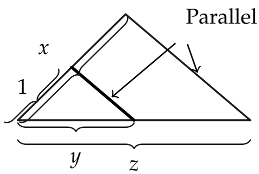

FIGURE 9–14

9–4 PROPORTION

Before discussing similar triangles, we recall some facts about proportions that you learned in arithmetic and algebra.

Definition 9–1. If x and y are numbers, with $y \neq 0$, the ratio of x to y is the number x/y, that is, x divided by y. A *proportion* is a statement that two (or more) ratios are equal. For example, $a/b = c/d$ is called a proportion. In this proportion, we say that a and c are proportional to b and d, respectively, and also that "a is to b as c is to d." More generally, the statement that x, y, z, u, … are proportional to x', y', z', u', … means that $x/x' = y/y' = z/z' = u/u' = \cdots$. In all proportions, we must require that the denominators are not zero, because division by zero is impossible.

You may recall that in algebra you learned a number of theorems about proportions, some of which are troublesome to remember. All statements about proportions are simple consequences of the basic laws of algebra. The theorems on proportions in this chapter are not to be memorized, but rather their proofs are to be understood.

Exercise Group 9–4

1. Give other ratios equal to the following:
(a) $9/12$, (b) $2x^2/3x$, (c) $12abc/27c^2$, (d) $(x^2 - x - 6)/(4x^2 + 8x)$, (e) $(x^2 - 5x + 6)/(x^2 - 4)$.

2. Are the following proportions true?
(a) $\dfrac{a}{b} = \dfrac{a^2}{ab}$, (b) $\dfrac{x}{y} = \dfrac{x^2}{y^2}$, (c) $\dfrac{3}{4} = \dfrac{9}{16}$, (d) $x = \dfrac{6x}{6}$.

3. Any number can be regarded as a ratio because $N = N/1$. Hence, can $x + 2$ be regarded as a ratio for

every number x?

4. Determine x, so that the following proportions are true: (a) $\frac{2x}{8} = \frac{5}{3}$, (b) $\frac{(3-x)}{x} = \frac{5}{3}$, (c) $\frac{2x+3}{5} = \frac{x-3}{2}$, (d) $\frac{ax}{5} = \frac{a}{10}$, *(e) $\frac{2x-7}{2} = \frac{3x-5}{3}$, (f) $\frac{9}{x} = \frac{24}{8}$, (g) $\frac{3}{x} = \frac{x}{12}$, (h) $\frac{a}{x} = \frac{x}{b}$.

5. Determine the ratio of u to v if (a) $4u = 5v$, (b) $v = 3u$, (c) $au = bv$, (d) $u = 3v$, (e) $4u - 18v = 0$, (f) $(2u - v)/(u+v) = \frac{1}{2}$.

6. If x is to 5 as 5 is to x, find x.

We list below some theorems about proportions that are simple consequences of the laws of algebra. Since a ratio is a number obtained by division, we first recall the meaning of division.

Definition 9–2. If $b \neq 0$, then a/b (or $a \div b$) is the number q, such that $a = qb$. In other words, for $b \neq 0$, $a/b = q$ if and only if $a = qb$.

Theorem 9–2. *If $b \neq 0$, and $c \neq 0$, then $a/b = (ac/bc)$.*

Proof. If $q = a/b$, then $a = qb$. Therefore, $ac = q \cdot bc$. Whence $q = (ac/bc)$.

Theorem 9–3. *If neither b nor $d = 0$, $a/b = c/d$ if and only if $ad = bc$.*

Proof. The statement $a/b = c/d$ says that two numbers are equal, that is, they are the same number. Multiplying this number by bd, we have $bd \cdot a/b = bd \cdot c/d$, or $da = bc$. Conversely, if $ad = bc$ and $bd \neq 0$, then dividing this number by bd, we have $(ad/bd) = (bc/bd)$ or $a/b = c/d$.

Theorem 9–4. *If $c \neq 0$, then $a/b = c/d$ if and only if $a/c = b/d$.*

Proof. Both proportions are equivalent to $ad = bc$, by Theorem 9–3.

Exercise Group 9–5

1. If $dx = ey$, what is the ratio of x to y?

2. If $3u = 7v$, what is the ratio of u to v? of v to u?

3. If $hx - ky = ry + ty$, what is the ratio of x to y?

*4. Prove that if $x/y = u/v$, $x = u$, and $x \neq 0$, then $v = y$.

*5. Prove that if $a/b = c/d$, and $b + d \neq 0$, then $a/b = (a+c)/(b+d)$.

*6. Prove that if $a/b = c/d$, then $(a+b)/b = (c+d)/d$.

7. Use Exercise 6 to form a new proportion from $x/3 = z/7$.

8. Use Exercise 5 to form a new proportion from $6/15 = 10/25$.

*9. State and prove the converse of Exercise 5.

*10. State and prove the converse of Exercise 6.

9–5 RATIO AND PROPORTION IN GEOMETRY

The objects of study in geometry are sets of points. Some of these objects have numbers attached to them; for example, a segment has a *length*. Later, we also discuss area of polygons and measurement of angles. In geometry, when we speak of the ratio of two segments, we mean the *ratio of the numbers* that are the lengths of the segments. It will be much the same when we speak of the ratio of any other geometric objects.

Definition 9–3. If AB and CD are segments, the ratio of AB to CD is the number $\overline{AB}/\overline{CD}$. The statement that the segments AB

and CD are *proportional* to the segments $A'B'$ and $C'D'$ means that $\overline{AB}/\overline{A'B'} = \overline{CD}/\overline{C'D'}$. We say that the sides of $\triangle ABC$ are proportional to the sides of $\triangle A'B'C'$ if $\overline{AB}/\overline{A'B'} = \overline{BC}/\overline{B'C'} = \overline{CA}/\overline{C'A'}$.

Exercise Group 9–6

1. Draw two segments. By measurement, determine the ratio of one segment to the other.

2. In Fig. 9–15, the sides of $\triangle EFG$ are proportional to the sides of $\triangle E'F'G'$. What is the ratio of EG to $E'G'$, if $\overline{E'F'} = 2$, and the lengths of the sides of $\triangle EFG$ are as shown? What is the ratio of $F'G'$ to $E'F'$?

3. If $\overline{AB} = 5$, $\overline{CD} = 7$, and $\overline{EF} = 3$, what is the ratio of AB to CD? of CD to EF? of AB to FE?

4. The sides of $\triangle ABC$ are proportional to the sides of $\triangle A'B'C'$ and $\overline{AB} = 5$, $\overline{BC} = 6$, $\overline{CA} = 8$. If $\overline{C'A'} = 9$, find $\overline{A'B'}$.

5. If the sides of $\triangle EFG$ are proportional to the sides of $\triangle PQR$ and $\overline{EF} = 7$, $\overline{FG} = 8$, $\overline{QR} = 6$, $\overline{RP} = 8$, find $\overline{GE}$ and $\overline{PQ}$.

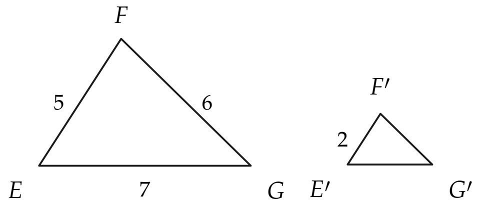

FIGURE 9–15

Before proceeding, we should observe that a question might be raised concerning Definition 9–3. The number that we get for the length of a segment depends on what we choose for a segment of unit length. Thus a foot ruler has length 1 if the unit length is one foot, but has length 12 if the unit is one inch. We now justify Definition 9–3 by proving that the ratio does not depend on the unit of length.

Theorem 9–5. *No matter what unit of length is chosen for measuring two segments, the ratio of these segments is the same number.*

Proof. Suppose that the two segments are AB and $A'B'$, and suppose that we have two different unit segments CD and EF, as in Fig. 9–16.

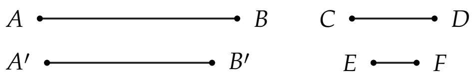

FIGURE 9–16

Suppose the lengths of AB and $A'B'$, when CD is the unit, are l and l' respectively, and their lengths, when EF is the unit, are L and L' respectively. We have seen, in Theorem 5–4, that changing the unit of measure just multiplies the length by a number. That is, if the length of CD, when EF is the unit, is x, then $L = l \cdot x$, and $L' = l' \cdot x$. Hence, if we form the ratio of AB to $A'B'$ when CD is the unit, we get l/l'; and if we form the ratio when EF is the unit, we get

$$\frac{L}{L'} = \frac{l \cdot x}{l' \cdot x} = \frac{l}{l'}.$$

Hence, the two ratios are the same, and the theorem is proved.

Exercise Group 9–7

1. Write the ratios above when CD is twice as long as EF; three times as long.
2. One inch is approximately 2.54 centimeters. What is the length of a foot rule in centimeters?
3. Two sides of a triangle have lengths 12 in. and 16 in. What would their ratio be if the sides were measured in miles?
4. If the ratio of segment AB to segment CD is 0.75, what is the ratio of CD to AB?
5. Is the ratio of one segment to another ever the number zero?

We have defined a ratio to be a number, and numbers can be either *rational* or *irrational*. When we encounter these two kinds of ratios in geometry, we use the words *commensurable* and *incommensurable*. We inherit this language from the Greeks, who discovered irrational numbers and were much disturbed by their discovery. We make the language clear in the following definition.

Definition 9–4. One segment is said to be *commensurable* with another if the ratio of the first to the second is a rational number. They are said to be *incommensurable* if this ratio is an irrational number.

Theorem 9–6. *Two segments, AB and CD, are commensurable with each other if and only if there is a segment EF which, if used as a unit of measure, makes $\overline{AB}$ and $\overline{CD}$ whole numbers (integers).*

Remark. Such a segment EF is a common unit of measure for the two segments, hence the word *commensurable*.

Proof. We have to show that if such a segment EF exists, then $\overline{AB}/\overline{CD}$ is a rational number, and conversely, that if $\overline{AB}/\overline{CD}$ is a rational number, then such a segment EF exists.

Suppose first that such a segment EF exists. Then $\overline{AB} = m$, and $\overline{CD} = n$, where m and n are integers. Therefore $\overline{AB}/\overline{CD} = m/n$, and is a rational number.

To prove the converse, suppose that $\overline{AB}/\overline{CD}$ is a rational number, that is, $\overline{AB}/\overline{CD} = m/n$, where m and n are integers. Divide the segment CD into n equal subintervals, and call one of these subintervals EF (Fig. 9–17). With EF as a unit of measure, $\overline{CD} = n$, and $\overline{AB} = m$.

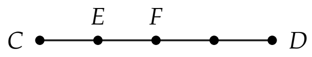

FIGURE 9–17

Note. In this book, our theorems deal only with commensurable pairs of segments, and we assume their truth for incommensurable cases. Proofs for incommensurable cases require some knowledge of *limits*. The ideas involved are difficult and we avoid limits whenever possible.

Historical remark. Pythagoras (6th century B.C.) believed that all quantitative relations could be described by pure numbers. By this, he meant whole numbers, or integers. He was (so the story goes) profoundly shocked when he discovered that the side and diagonal of a square are incommensurable. Because of this discovery, later Greeks (notably Eudoxus and Euclid) invented an elaborate theory of ratio and proportion that is a model of ingenuity and precision. The interesting fact about this theory, from our point of view, is that *there were no numbers in Greek geometry, that is, no measurement*, and hence the meaning of the words “two ratios are equal” had to be defined in a special way that we cannot consider here.

9–6 SIMILAR TRIANGLES AND POLYGONS

In this section we are primarily concerned with similar triangles, but we also define similarity for polygons, and solve some problems concerning pairs of similar polygons. Remember that we are trying to make precise what we mean when we say that two polygons *have the same shape*.

Definition 9–5. Two *triangles* are said to be *similar* to each other if corresponding angles are congruent and corresponding sides are proportional. For example, in Fig. 9–18, $\triangle ABC$ is similar to $\triangle A'B'C'$ if $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, $\angle C \cong \angle C'$, and $\overline{AB}/\overline{A'B'} = \overline{BC}/\overline{B'C'} = \overline{CA}/\overline{C'A'}$. To indicate that triangles are similar, we write $\triangle ABC \sim \triangle A'B'C'$.

Two *convex* polygons are said to be *similar* to each other if the corresponding angles are congruent and the corresponding sides are proportional. For example, the polygons in Fig. 9–19 are similar if $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, $\angle C \cong \angle C'$, $\angle D \cong \angle D'$, $\angle E \cong \angle E'$, and $\overline{AB}/\overline{A'B'} = \overline{BC}/\overline{B'C'} = \overline{CD}/\overline{C'D'} = \overline{DE}/\overline{D'E'} = \overline{EA}/\overline{E'A'}$.

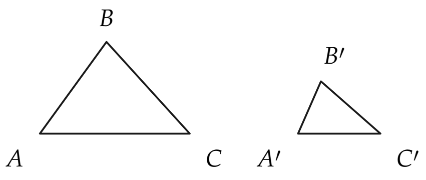

FIGURE 9–18

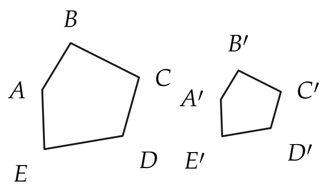

FIGURE 9–19

Exercise Group 9–8

1. Two triangles are similar, and two sides of one triangle are 7 and 11 in. In the other triangle, the side corresponding to the 7-in. side has length 12 in. Find the length of the side corresponding to the 11-in. side.

2. Two triangles are similar, and the sides of one triangle have lengths 4, 6, and 8. In the other triangle, the shortest side has length 6. Find the lengths of the other sides.

3. If $\triangle ABC \sim \triangle DEF$ and $\overline{AC} = 21$, $\overline{DF} = 15$, $\overline{FE} = 18$, and $\overline{DE} = 21$, find the lengths of the other sides of $\triangle ABC$.

4. Two quadrilaterals are similar, and one quadrilateral has sides of 5, 7, 12, and 16 in. The other quadrilateral has its longest side 12 ft. How long are the other sides?

5. The sides of a triangle are 4, 5, and x in. The corresponding sides of a triangle similar to the first triangle are 6, y, and 8 in. What are the sides of each triangle?

6. A pentagon has sides 4, 6, 10, 15, and x ft. In a pentagon similar to the first pentagon, the corresponding sides are u, v, 8, w, and 16. Find u, v, w, and x.

7. Two right triangles are similar, and in the first, the ratio of the shorter to the longer leg is $\frac{2}{3}$. In the other triangle, the shorter leg is 13 in. long. How long is the longer leg of this triangle?

8. Find as many examples of similar polygons from the everyday world as you can.

9. Prove that if $\triangle ABC \sim \triangle DEF$ and $\triangle DEF \sim \triangle PQR$, then $\triangle ABC \sim \triangle PQR$.

10. Are all squares similar? Why?

11. Are all rectangles similar? Why?

12. State conditions under which two parallelograms are similar.

13. Prove that congruent polygons are similar polygons.

Is the converse true?

14. Prove that all equilateral triangles are similar.

15. In Fig. 9–20, E and F are midpoints of AB and BC. Prove that $\triangle BEF \sim \triangle BAC$.

16. If, in Fig. 9–21, A', B', C', and D' bisect the corresponding segments from O, prove that the quadrilaterals $ABCD$ and $A'B'C'D'$ are similar.

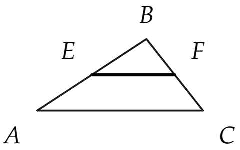

FIGURE 9–20

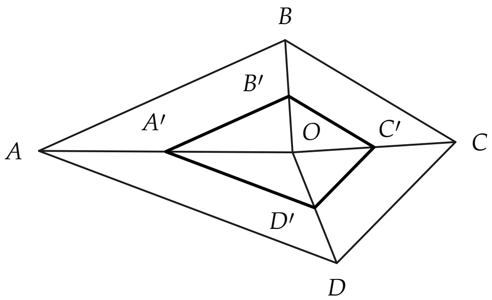

FIGURE 9–21

9–7 THEOREMS ON SIMILAR TRIANGLES

In this section we give the fundamental theorems on similar triangles. They are extremely important because they have wide application in mathematics, other sciences, and engineering.

Theorem 9–7. *If the angles of one triangle are congruent, respectively, to the angles of another triangle, then the triangles are similar.*

Proof. Suppose that the triangles are $\triangle ABC$ and $\triangle A'B'C'$, as in Fig. 9–22, and that $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, and $\angle C \cong \angle C'$. We must prove that the sides are proportional, that is, $\overline{AB}/\overline{A'B'} = \overline{BC}/\overline{B'C'} = \overline{CA}/\overline{C'A'}$.

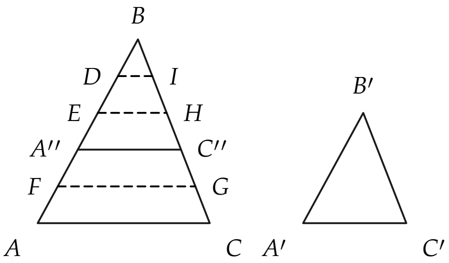

FIGURE 9–22

In forming the proof, we will suppose that the segments AB and $A'B'$ are commensurable. In other words, $\overline{AB}/\overline{A'B'}$ is a rational number. The proof for the incommensurable case is difficult, since it involves knowing about *limits*. We therefore suppose that, with a suitable unit, the lengths of AB and $A'B'$ are positive integers. The figure is drawn with $\overline{AB} = 5$ and $\overline{A'B'} = 3$. We may suppose that $AB > A'B'$, for if $AB < A'B'$, the argument can be reversed, and if $AB \cong A'B'$, the triangles are congruent (why?), and hence similar.

STATEMENTS	REASONS
1. There is a point A'' on AB, such that $\overline{A''B} \cong \overline{A'B'}$, and $\overline{A''B} = \overline{A'B'} = 3$.	Congruence postulate.
2. A'' is between A and B.	We have assumed that $AB > A'B'$.
3. There is a line through A'' parallel to AC which intersects BC in C''.	Why?
4. $\triangle BA''C'' \cong \triangle B'A'C'$.	A.S.A.
5. $\overline{BC''} \cong \overline{B'C'}$.	Statement 4.
6. There are points D, E, and F, on AB, such that the points A, F, A'', E, D, and B, in that order, are at unit distance from their neighbors.	It is given that $\overline{AB} = 5$. By Statement 1, $\overline{A''B} = \overline{A'B'} = 3$.
7. There are lines through D, E, and F, parallel to AC. These lines intersect BC in points I, H, and G, respectively.	Why?
8. $\overline{CG} \cong \overline{GC''} \cong \overline{C''H} \cong \overline{HI} \cong \overline{IB}$.	Theorem 9–1 on transversals.
9. If CG is chosen as unit of length, $\overline{BC} = 5$, and $\overline{BC''} = \overline{B'C'} = 3$.	Definition of length and Statement 5.
10. $\overline{BC}/\overline{B'C'} = 5/3$.	Statement 9.
11. $\overline{AB}/\overline{A'B'} = 5/3$.	Given.
12. $\overline{AB}/\overline{A'B'} = \overline{BC}/\overline{B'C'}$.	Statements 10 and 11.
13. In an analogous way, by considering a line parallel to BC, it may be shown that $\overline{AB}/\overline{A'B'} = \overline{AC}/\overline{A'C'}$.	Same reasons as in Statements 1 through 12.
14. $\triangle ABC \sim \triangle A'B'C'$.	Statements 12 and 13.

Remark. It should be clear that the proof is essentially the same if AB and $A'B'$ are any other positive integers besides 5 and 3.

Corollary 9–7–1. *If two angles of one triangle are congruent to two angles of another triangle, then the triangles are similar.*

Corollary 9–7–2. *If an acute angle of one right triangle is congruent to an acute angle of another right triangle, then the triangles are similar.*

The proofs are left to the student.

Exercise Group 9–9

1. If, in Fig. 9–23, $BE \parallel CD$, $\overline{AE} = 4$, and $\overline{ED} = 6$, what is $\overline{AB}/\overline{AC}$?

2. A church steeple casts a shadow 100 ft long, and a 6-ft post casts, at the same time, a shadow 7 ft long. How high is the steeple?

3. A line from the top of a cliff to the ground just passes over the top of a pole 8 ft high and meets the ground at a point 5 ft from the base of the pole. If it is 60 ft from this point to the base of the cliff, how high is the cliff?

4. As in Fig. 9–24, a lake lies between A and C. If lines AB and DE run north-south, $\overline{AB} = 8$ mi, $\overline{DE} = 1.2$ mi, and $\overline{DC} = 1.5$ mi, how far is it from A to C?

5. In Fig. 9–25, $DE \parallel AC$. Prove that $\triangle ABC \sim \triangle DBE$.

6. If in $\triangle DEF$, G and H are points on sides EF and DF, respectively, and $\angle GHF \cong \angle EDF$, prove that $\triangle DEF \sim \triangle HGF$.

7. In Fig. 9–26, $AB \perp AE$, and $DE \perp AE$. Prove that (a) $\overline{AB}/\overline{DE} = \overline{AC}/\overline{CE}$; (b) $\overline{AB}/\overline{BC} = \overline{DE}/\overline{DC}$.

8. In $\triangle ABC$, let E, F, and G be the midpoints of the sides AB, BC, and CA, respec-

tively. Prove that $\triangle ABC \sim \triangle FGE$.

9. If $\triangle ABC \sim \triangle A'B'C'$, and BD and $B'D'$ are the altitudes from B and B' respectively, prove that $\overline{AC}/\overline{A'C'} = \overline{BD}/\overline{B'D'}$.

10. In Fig. 9–27, $l \parallel m$. Prove that $\triangle EFG \sim \triangle IHG$.

11. If, in Fig. 9–28, $\triangle PQR \sim \triangle P'Q'R'$, and PS and $P'S'$ bisect angles P and P', prove that $\overline{PQ}/\overline{PS} = \overline{P'Q'}/\overline{P'S'}$.

12. In $\triangle ABC$, CD and AE are altitudes, as in Fig. 9–29. Prove that (a) $\triangle BEA \sim \triangle BDC$; (b) $\triangle ADF \sim \triangle CEF$.

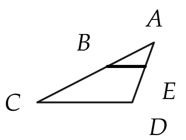

FIGURE 9–23

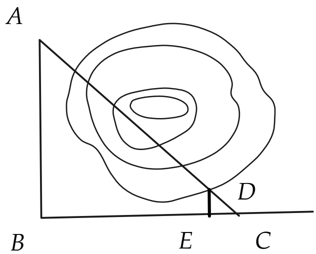

FIGURE 9–24

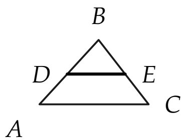

FIGURE 9–25

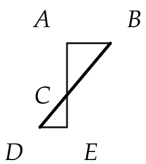

FIGURE 9–26

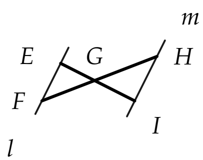

FIGURE 9–27

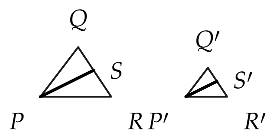

FIGURE 9–28

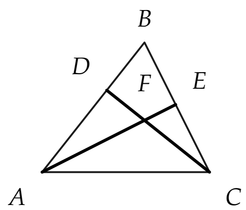

FIGURE 9–29

Before proving the next of the fundamental theorems on similar triangles, we prove a theorem that we need later and which is interesting in itself.

Theorem 9–8. *A line that divides two sides of a triangle proportionally is parallel to the third side. This means that if* $\overline{AD}/\overline{AB} = \overline{AE}/\overline{AC}$ *in Fig. 9–30, then* $DE \parallel BC$.

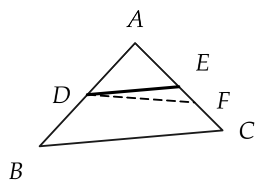

FIGURE 9–30

Proof.

STATEMENTS	REASONS
1. Let DF be a line parallel to BC and meeting AC in a point F.	Why is there such a line?
2. The point F is on the segment AC.	Why?
3. $\triangle ABC \sim \triangle ADF$.	Explain how Theorem 9–7 is used here.
4. $\overline{AF}/\overline{AC} = \overline{AD}/\overline{AB}$.	From the definition of similar triangles.
5. $\overline{AD}/\overline{AB} = \overline{AE}/\overline{AC}$.	Given.
6. $\overline{AF} = \overline{AE}$.	Statements 4 and 5.
7. $F = E$.	$AF \cong AE$ and congruence postulates.
8. $DE \parallel BC$.	Statements 7 and 1.

Theorem 9–9. *Given $\triangle ABC$ and $\triangle A'B'C'$ with $\angle A \cong \angle A'$ and $\overline{AB}/\overline{A'B'} = \overline{AC}/\overline{A'C'}$, then $\triangle ABC \sim \triangle A'B'C'$.*

Proof. We show that $\angle B \cong \angle B'$, and use Corollary 9–7–1. We may suppose that $AB > A'B'$. (Why?)

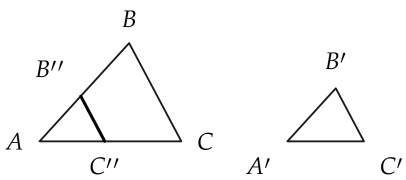

FIGURE 9–31

STATEMENTS	REASONS
1. In Fig. 9–31, there are points B'' on AB and C'' on AC such that $\overline{A'B'} \cong \overline{AB''}$ and $\overline{A'C'} \cong \overline{AC''}$.	Why?
2. $\triangle AB''C'' \cong \triangle A'B'C'$.	Why?
3. $\overline{AB}/\overline{AB''} = \overline{AC}/\overline{AC''}$.	Why?
4. $B''C'' \parallel BC$.	Theorem 9–8.
5. $\angle AB''C'' \cong \angle B$.	Why?
6. $\angle B \cong \angle B'$.	Statements 2 and 5.
7. $\triangle ABC \sim \triangle A'B'C'$.	Why?

Exercise Group 9–10

1. Prove that if the legs of one right triangle are proportional to the legs of another right triangle, then the triangles are similar.

2. In Fig. 9–32, $\overline{AO} = 10$, $\overline{BO} = 12$, $\overline{OD} = 5$, and $\overline{OC} = 6$. Show that $\triangle AOB \sim \triangle DOC$.

3. Prove that two isosceles triangles are similar if their vertex angles are congruent; likewise if their base angles are all congruent to one another.

4. In Fig. 9–33, if $\overline{AD} = x \cdot \overline{AB}$, and $\overline{AE} = x \cdot \overline{AC}$, prove that $\triangle ABC \sim \triangle ADE$.

5. If $\triangle ABC$ is a right triangle, with the right angle at C, and if, in Fig. 9–34, $\overline{AD}/\overline{AC} = \overline{AC}/\overline{AB}$, prove that $CD \perp AB$.

*6. In Fig. 9–35, if $\triangle ABC$ is a right triangle, with right angle at C, and if $CD \perp AB$, prove that $\triangle ABC \sim \triangle ACD$.

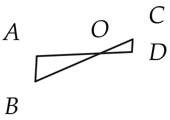

FIGURE 9–32

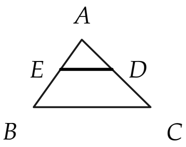

FIGURE 9–33

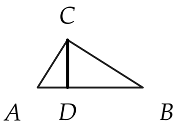

FIGURE 9–34

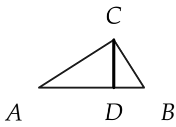

FIGURE 9–35

The next theorem is the last of the three basic theorems on similar triangles.

Theorem 9–10. *If the sides of one triangle are proportional to the sides of another triangle, then the triangles are similar. In other words, referring to Fig. 9–36, if* $\overline{AB}/\overline{A'B'} = \overline{AC}/\overline{A'C'} = \overline{BC}/\overline{B'C'}$, *then* $\triangle ABC \sim \triangle A'B'C'$.

Proof. It is sufficient to show that $\angle B \cong \angle B'$. Why? We may assume that $AB > A'B'$, for if $AB < A'B'$, the argument can be reversed, and if $\overline{AB} = \overline{A'B'}$, then the triangles are congruent (why?), and hence similar.

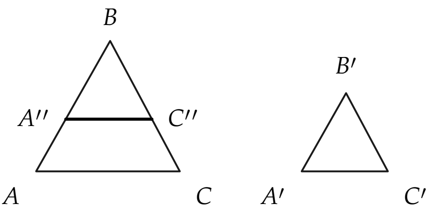

FIGURE 9–36

STATEMENTS	REASONS
1. Choose A'' on AB and C'' on BC, such that $\overline{A''B} = \overline{A'B'}$ and $\overline{C''B} = \overline{C'B'}$.	Why possible?
2. $\overline{AB}/\overline{A''B} = \overline{BC}/\overline{BC''}$.	The hypothesis of Theorem 9–10 and Statement 1.
3. $\triangle ABC \sim \triangle A''BC''$.	Theorem 9–9.
4. $\overline{AC}/\overline{A''C''} = \overline{AB}/\overline{A''B} = \overline{AB}/\overline{A'B'}$.	Statements 3 and 1.
5. $\overline{AC}/\overline{A''C''} = \overline{AC}/\overline{A'C'}$.	Statement 4 and the given fact $\overline{AB}/\overline{A'B'} = \overline{AC}/\overline{A'C'}$.
6. $\overline{A''C''} = \overline{A'C'}$.	Statement 5.
7. $\triangle A''BC'' \cong \triangle A'B'C'$.	Statements 1 and 6 and S.S.S.
8. $\angle B \cong \angle B'$.	Statement 7.
9. $\triangle ABC \sim \triangle A'B'C'$.	Theorem 9–9.

Exercise Group 9–11

1. State, in your own words, the three basic theorems on similar triangles, Theorems 9–7, 9–9, and 9–10.

2. In Fig. 9–37, the numbers are the lengths of the segments. How are the lines AB and CD related? Why?

3. If the hexagons of Fig. 9–38 are similar, prove that (a) $\triangle BCD \sim \triangle B'C'D'$; (b) quadrilateral $ABCD \sim$ quadrilateral $A'B'C'D'$.

4. In Fig. 9–39, $BE \perp AC$, and $CD \perp AB$. Find two similar triangles. How many proportions can you write?

5. If $\triangle ABC \sim \triangle A'B'C'$, and AD and $A'D'$ are medians, prove that the ratio of the medians is equal to the ratio of any two correspond-

ing sides. Restate this using the word *proportion* (or *proportional*).

6. Prove that if two triangles are similar, then the ratio of their perimeters is the same as the ratio of two corresponding sides.

7. Is a statement like that of Exercise 6 true for similar polygons? How would you prove this?

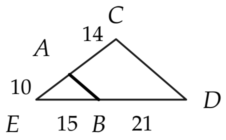

FIGURE 9–37

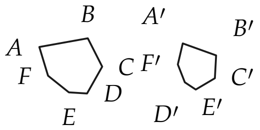

FIGURE 9–38

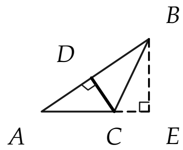

FIGURE 9–39

9–8 THE PYTHAGOREAN THEOREM

With our knowledge of similar triangles, we are able to prove a theorem associated with the Greek mathematician Pythagoras, who lived in the 6th century B.C. It is believed that he first gave a proof of the theorem that bears his name. First, we state a theorem that is useful for the proof of the Pythagorean theorem.

Theorem 9–11. *The altitude to the hypotenuse of a right triangle forms two right triangles, which are similar to the given triangle. For example, in Fig. 9–40,* $\triangle ABC \sim \triangle ACD \sim \triangle CBD$.

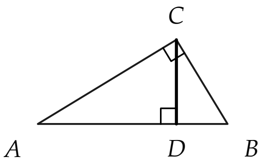

FIGURE 9–40

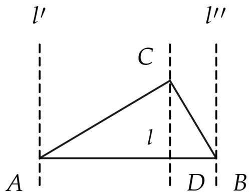

FIGURE 9–41

Proof. In the first place, a line through C perpendicular to AB must intersect the segment AB. This fact may be seen as follows: Consider lines l, l', and l'' in Fig. 9–41, through C, A, and B, and perpendicular to AB. Then $l' \parallel l \parallel l''$. The point C is between l' and l'', because angles A and B are acute angles. Therefore, l is between l' and l'', and hence l intersects AB in a point D. $\triangle ACD \sim \triangle ABC$, since each triangle has a right angle, and the triangles have the acute angle A in common. By the same reasoning, $\triangle BCD \sim \triangle BAC$.

Corollary 9–11–1. *The altitude to the hypotenuse is a mean proportional* between the two segments formed on the hypotenuse.*

Proof. We have to show that, using Fig. 9–40, $\overline{AD}/\overline{CD} = \overline{CD}/\overline{DB}$. Since $\triangle ADC \sim \triangle CDB$, this is an immediate consequence of the definition of similar triangles.

Corollary 9–11–2. *A leg of a right triangle is the mean proportional between the hypotenuse and the adjacent segment on the hypotenuse formed by the altitude from the right angle. That is, in Fig. 9–40, $\overline{AC}^2 = \overline{AD} \cdot \overline{AB}$.*

Exercise Group 9–12

1. In Fig. 9–42, $EF \perp FG$, and $FH \perp EG$. If $\overline{EF} = 8$, and $\overline{FG} = 5$, what is $\overline{FH}/\overline{EH}$? $\overline{FH}/\overline{HG}$?

2. In Fig. 9–43, the sides of a right triangle are 5, 12, and 13. How long is the altitude on the hypotenuse?

3. In Fig. 9–44, Q is a right angle, and $QS \perp PR$. Find $\overline{PQ}$, if $\overline{PS} = 4$, and $\overline{SR} = 5$.

*x is a "mean proportional" between a and b if $a/x = x/b$, or, which is the same, $x^2 = ab$.

4. Using Fig. 9–44, if $\overline{QS} = 7$, what is the product $\overline{PS} \cdot \overline{SR}$?

5. In Fig. 9–45, $\angle C$ is a right angle, as is $\angle ADC$. Express h in terms of lengths a, b, and c of the sides. Obtain $\overline{AD}$ in terms of a, b, and c.

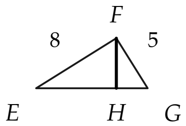

FIGURE 9–42

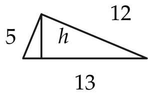

FIGURE 9–43

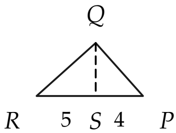

FIGURE 9–44

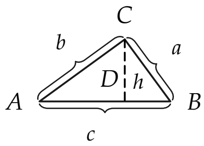

FIGURE 9–45

Theorem 9–12 (Pythagorean theorem). *The square of the length of the hypotenuse of a right triangle is equal to the sum of the squares of the lengths of the legs.*

If the triangle is $\triangle ABC$, with the right angle at C, and if the lengths of the sides opposite angles A, B, and C are a, b, and c, respectively, as in Fig. 9–46, then we are able to show that $c^2 = a^2 + b^2$.

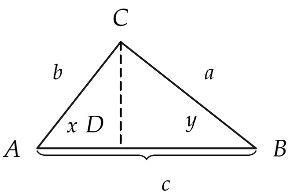

FIGURE 9–46

Proof.

STATEMENTS	REASONS
1. Let CD be the altitude and $x = \overline{AD}$ and $y = \overline{DB}$. Then $x/b = b/c$ and $y/a = a/c$.	Corollary 9–11–2.
2. $x + y = c$.	Properties of length.
3. $b^2/c + a^2/c = c$.	Statements 1 and 2.
4. $a^2 + b^2 = c^2$.	Statement 3 and algebra.

Theorem 9–13 (Converse to Pythagorean theorem). *If a triangle has sides of length a, b, and c, such that $c^2 = a^2 + b^2$, then the triangle is a right triangle, and the side of length c is the hypotenuse.*

Proof. The method of proof is to show that there is a right triangle that is congruent to the given triangle. Suppose that the given triangle is $\triangle ABC$.

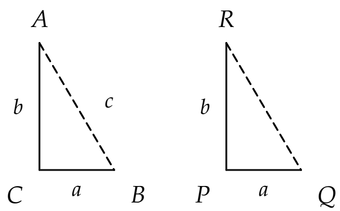

FIGURE 9–47

STATEMENTS	REASONS
1. There is a segment $PQ \cong BC$, as in Fig. 9–47.	Congruence postulates.
2. There is a ray perpendicular to PQ at P.	From the congruence postulates, there is an angle at P congruent to a right angle.
3. There is a point R on this ray such that $PR \cong AC$.	Congruence postulates.
4. $\overline{PQ} = a$, and $\overline{PR} = b$.	Statements 1 and 3.
5. $\overline{RQ}^2 = a^2 + b^2$.	Pythagorean theorem.
6. $c^2 = a^2 + b^2$.	Given.
7. $\overline{RQ}^2 = c^2$, so $\overline{RQ} = c$.	Statements 5 and 6 and algebra.
8. $RP \cong AC$, $PQ \cong CB$, and $RQ \cong AB$.	By construction and Statement 7.
9. $\triangle ABC \cong \triangle RQP$.	S.S.S.
10. $\angle C$ is a right angle.	$\angle P$ is a right angle, by construction, and $\angle C \cong \angle P$, by Statement 9.

Exercise Group 9–13

1. It is said that the ancient Egyptians constructed right angles for surveying purposes by forming a triangle using ropes of length 3, 4, and 5 units. Why would this be correct? Can you think of other integers for the sides of a right triangle?

2. In a right triangle, the hypotenuse is 7 and one leg is 3. How long is the other leg?

3. Write a formula for the length of one leg of a right triangle in terms of the

lengths of the hypotenuse and other leg.

*4. In Fig. 9–48, $AB \parallel CD$, $BC \parallel AD$, and $AC \perp BD$. If $\overline{BD} = 24$, and $\overline{AD} = 13$, find $\overline{AC}$.

5. In a right triangle, the two legs are 6 ft and 9 ft. How long is the hypotenuse?

6. The two legs of a right triangle are equal, and the hypotenuse is 4 in. long. How long are the legs?

7. The base of an isosceles triangle is 12 in. long. The two congruent sides have a combined length of 13 in. How long is the altitude to the base?

8. Point A in Fig. 9–49 is 21 mi due north of O, while point B is 20 mi due east of O. What is the distance from A to B, across the lake shown?

9. Find the altitude of an equilateral triangle, the length of whose side is a.

10. A rope 100 ft long reaches from the top of a pole to a point 40 ft from the foot of the pole. How tall is the pole?

11. Prove that the median to the hypotenuse of a right triangle is half as long as the hypotenuse.

12. Prove that in a 30°–60° right triangle, the shorter leg is half as long as the hypotenuse.

13. Prove that in an isosceles right triangle, the ratio of leg to hypotenuse is $1/\sqrt{2}$.

14. Construct a segment of length $\sqrt{5}$ in.

15. What is the length of a diagonal of a square of side 1? of side a?

16. Construct a segment of length $\sqrt{6}$.

*17. If n is any positive integer, can you construct (RC) a segment of length $\sqrt{n}$?

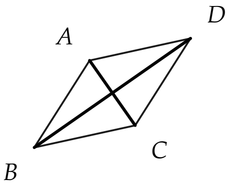

FIGURE 9–48

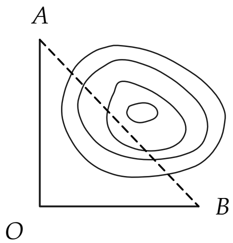

FIGURE 9–49

9–9 ON CONSTRUCTIONS

In our previous study of RC constructions, we assumed that two circles with equal radii would intersect if the distance between their centers were less than the sum of their radii. We can easily prove this by using the Pythagorean theorem. To do this, we first calculate where they *should* intersect, and then prove that this point *is* on both circles.

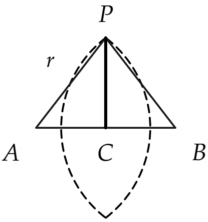

FIGURE 9–50

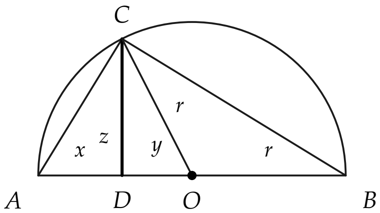

FIGURE 9–51

Consider Fig. 9–50, where r is the radius of the circles, and $d = \overline{AB}$ is the distance between their centers. If the circles intersect at P, and C is the midpoint of AB, then $\triangle PAC \cong \triangle PBC$ (why?), and hence $\angle C$ is a right angle. Therefore, $r^2 = \overline{PC}^2 + \overline{AC}^2$, so that $\overline{PC} = \sqrt{r^2 - (\frac{1}{2}\overline{AB})^2}$. We now know "where" the point of intersection P must be, and we can prove that such a point of intersection exists.

Theorem 9–14. *Suppose that circles, with centers A and B, have equal radii, $r > \frac{1}{2}AB$. These circles intersect.*

Proof. Let C be the midpoint of AB. Through C there is a line perpendicular to AB. On this line, let P be a point at the distance $\overline{PC} = \sqrt{r^2 - (\frac{1}{2}\overline{AB})^2}$ from C. (How do we know there is such a point P?) Because $\triangle PAC$ and $\triangle PBC$ are right triangles, it follows from the Pythagorean theorem that $\overline{PA} = \overline{PB} = r$. Hence P lies on both circles.

Remark. It also follows from the above proof that the two circles can intersect in two, and only two, points, for the following reason. On the line through C, perpendicular to AB, there are only two points at the correct distance from C, namely one on each side of the line AB.

Theorem 9–15. *An angle inscribed in a semicircle is a right angle. That is, if in Fig. 9–51, AB is a diameter of the circle, with center at O, and C is any point on the circle other than A and B, then $\angle ACB$ is a right angle.*

Proof outline. We drop a perpendicular from C to AB, and prove that the right triangles ADC and CDB are similar. It then follows that $\angle C$ is a right angle.

STATEMENTS	REASONS
1. Let CD be the perpendicular from C to AB, and suppose that D is between A and O.	The point D must be between A and B. (Why?) D, then, either is O, or is between A and O, or O and B. If $D = O$, the proof is easy. We discuss only the case in which D is between A and O. If D is between O and B, a similar proof can be made.
2. Let x, y, z, and r be the lengths of the segments in Fig. 9–51. Then $x = r - y$, and $z = \sqrt{r^2 - y^2}$.	Why?
3. $\dfrac{x}{z} = \dfrac{r-y}{\sqrt{r^2-y^2}} = \sqrt{\dfrac{r-y}{r+y}}$.	Statement 2 and algebra.
4. $\dfrac{z}{r+y} = \dfrac{\sqrt{r^2-y^2}}{r+y} = \sqrt{\dfrac{r-y}{r+y}}$.	Statement 2 and algebra.
5. $\triangle ACD$ and $\triangle CDB$ are right triangles.	By construction.
6. $\triangle ACD \sim \triangle CDB$.	Statements 3, 4, and 5 and Theorem 9–9.
7. $\angle ACD \cong \angle CBD$.	Statement 6.
8. $\angle ACB$ is a right angle.	Why?

Exercise Group 9–14

1. A circle passes through the vertices of a right triangle whose sides are 3, 4, and 5 in. What is the radius of the circle?

2. If PQ is a diameter of a circle of radius 13, and R is a point on the circle, such that $\overline{PR} = 10$, how long is $\overline{RQ}$?

3. In Fig. 9–52, AB is a diameter of the circle, and $CD \perp AB$. If $\overline{AB} = 30$ and $\overline{EO} = 12$, find $\overline{CD}$.

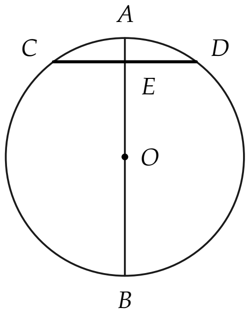

FIGURE 9–52

REVIEW OF CHAPTER 9

Be sure you know the definitions given in this chapter of the following concepts: transversal, ratio, proportion, ratio of segments, proportional sides, commensurable, similar triangles.

Some of the important theorems of the chapter were concerned with these definitions. Try to recall the theorems.

The main ideas of the chapter are listed briefly below. Be sure that you know them, and review the proofs.

1. If parallel lines cut off congruent segments on one transversal, they do so on every transversal.
   This theorem depended upon what facts about parallel lines and triangles?
   This theorem enables one to divide a segment into m congruent parts with ruler and compass. Show how to do this.

2. What is a proportion? What is the ratio of x to y if x and y are numbers? if x and y are segments?
   If neither b nor d is zero, $a/b = c/d$ if and only if $ad = bc$. Illustrate this theorem by numerical examples.

3. The ratio of two segments does not depend on the unit of length.
   Illustrate this fact by getting the ratio of depth to width of your desk, first measuring in feet and then in inches.
   On what theorem about measurement of length does this fact depend?

*4. Two segments are defined to be commensurable with each other if their ratio is a rational number. What is a rational number? Give an example of one. Give an example of an irrational number. What are incommensurable segments?
   State a theorem concerning commensurable segments.

5. What is the definition of similar triangles and polygons? Draw two similar triangles; pentagons.
   What do an architect's drawings have to do with similar figures?
   What does a map have to do with similarity?

6. Two triangles with corresponding angles congruent are similar.
   Illustrate the proof of this theorem, using two similar triangles ABC and $A'B'C'$, with $AB = 12$, and $A'B' = 8$.

7. If two triangles have their corresponding sides proportional, they are similar.
   What must be established to prove this theorem?

8. The Pythagorean theorem is proved, using a theorem about similar triangles in a right triangle. State this theorem.
   Prove the Pythagorean theorem.
   State its converse.

Review Exercises

1. $\overline{AB} = 24$, and C is a point on the line AB. If the ratio of AC to $CB = 3/5$, determine $\overline{AC}$ (a) if C is between A and B; (b) if A is between C and B.

2. Explain how to locate the point C in Exercise 1(a) by an RC construction.

*3. Can you give an RC construction for the point C in Exercise 1(b)?

4. A flagpole throws a 20-ft shadow when a yardstick throws a 10-in. shadow. Determine the height of the flagpole.

5. The length of the sides of a triangle are 4, 5, and 6. A triangle similar to this one has a perimeter of 30 in. Give the length of each side of this triangle.

6. In $\triangle ABC$, the altitude from vertex A divides the triangle into two similar triangles. $\triangle ABC$ is not isosceles. What conclusion may be drawn?

7. In $\triangle ABC$, $\angle A = 40°$, and $\angle B = 80°$. In $\triangle A'B'C'$, $\angle B' = 40°$, and $\angle C' = 60°$. Are the two triangles similar?

8. What is the mean proportional of 4 and 9? of 2 and 4? of 6 and 6?

9. A triangle whose sides have lengths 3, 4, and 5 is a right triangle. So also is the triangle with sides of 5, 12, and 13; 7, 24, and 25; 9, 40, and 41; 11, 60, and 61; etc. Explain the pattern underlying the formation of these triples of numbers.

10. On a certain map, a distance of $\frac{5}{8}$ in. represents 80 mi. On this map, the points representing two cities are $5\frac{7}{16}$ in. apart. What is the airline distance between these cities?

11. The parallel sides of a trapezoid have lengths 18 and 12, as in Fig. 9–53. The altitude is 8, and one of the nonparallel sides has length 10. If the nonparallel sides are extended, a triangle is formed by these extended sides and the greater base. (a) Show that this triangle is right. (b) Compute the lengths of the sides of this triangle.

12. If, in Exercise 11, the altitude is 6, rather than 8, and the other dimensions are as given, compute the length of the altitude of the triangle from the vertex opposite the side of length 18.

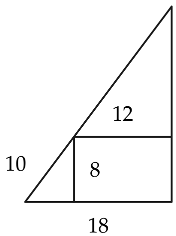

FIGURE 9–53

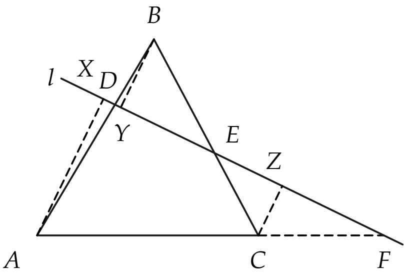

FIGURE 9–54

13. A parallelogram has adjacent sides of lengths 12 and 8, and these sides form a 60° angle. Find the lengths of the diagonals of the parallelogram. Find the lengths of the diagonals, if the included angle is 30°.

14. Draw three segments, and label them AB, CD, and EF. Locate by a ruler and compass construction a point X on AB, such that the ratio of AX to XB is equal to the ratio of CD to EF.

*15. In Fig. 9–54, line l cuts sides AB and BC of $\triangle ABC$ in points D and E, and cuts AC extended in point F. Prove that the product of the three ratios, AD to DB, BE to EC, and CF to

FA, is 1. [*Hint:* Consider perpendiculars AX, BY, and CZ from A, B, and C to l. Then $\overline{AD}/\overline{DB} = \overline{AX}/\overline{BY}$, etc.]

16. Construct a triangle similar to a given triangle and having a given perimeter.

17. Upon a given segment as side, construct a triangle similar to a given triangle.

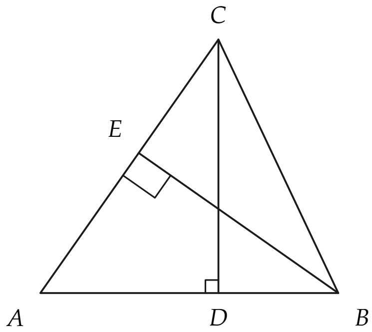

FIGURE 9–55

18. In $\triangle ABC$, CD and BE are altitudes, as in Fig. 9–55. Prove that $\overline{CD}/\overline{EB} = \overline{AC}/\overline{AB}$.

19. Prove that if, in triangles ABC and $A'B'C'$, $AB \parallel A'B'$, $BC \parallel B'C'$, and $CA \parallel C'A'$, then $\triangle ABC \sim \triangle A'B'C'$.

20. Prove that the line joining the midpoints of two opposite sides of a parallelogram is parallel to the two other sides and passes through the point of intersection of the diagonals.